

of temperature on the FPGA. The currents, temperatures and functionalities of the DB6-2 were stable during the full run. DB6-3 was irradiated in the ALC-144 cyclotron proton beam line with 54 MeV protons at the IFJ in Krakow to 381 Gy for half-A FPGA and 400 Gy for half-B FPGA at a rate of 2.88 Gy/h (figure 3). All the currents and the temperatures were stable during the full run. The higher fluctuations compared with DB6-1 (0 to 140 Gy) and DB6-2 are due to active Single Event Upset (SEU) corrections by the Xilinx SEM and resets applied when an uncorrectable SEU appeared during the run.

For the Single Event Latchups (SEL) tests, there was no prototype of the DB produced and the design was migrating from the 16 nm FinFET powered KU+ FPGA to the KU FPGA, due to the KU+ alternative been susceptible to SEL occurrences. As an alternative to the ProASIC, a Lattice ICE40LP FPGA was also tested [11]. Two separate tests were performed for SEL, one as described for the DB6-3 with 54 MeV protons, where the FPGAs were irradiated to a fluence of 2.62×10^{11} p/cm². In the second test, two Trenz TE0841 boards (each with the same KU FPGA used in the DB design), a Lattice ICE40LP FPGA and a ProASIC3E A3P31500 FPGA were irradiated with 226 MeV protons to a fluence of 1.11×10^{12} p/cm², for which all the currents of the FPGAs were monitored (figure 6 of ref. [11]). No SEL were observed in the units under test in either of the tests.

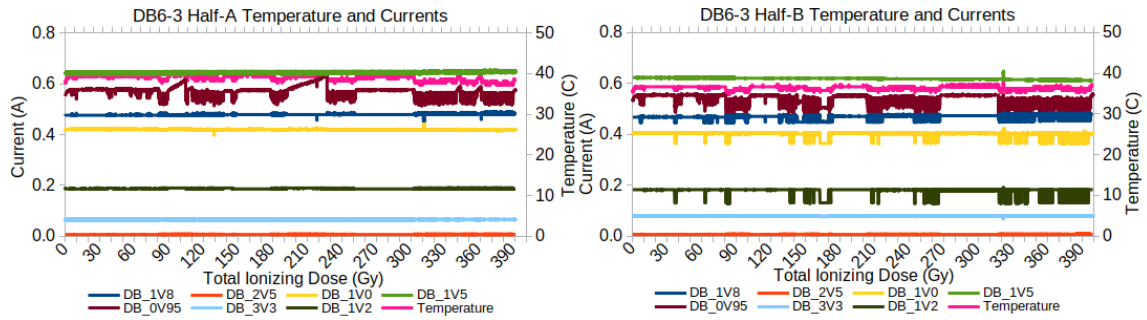


Figure 3. TID test of DB6-3. All board currents and the temperatures of both KU FPGAs were monitored over the whole irradiation period delivered by a 54 MeV proton beam.

A set of tests was arranged in coordination with the RIC Jozef Stefan Institute in Ljubljana, to cover the NIEL tolerance, where a DB (DB6-0) and a set of test boards with non-radiation qualified DB components were irradiated in the beam port number 6 of the TRIGA reactor [5] to a fluence of 14×10^{12} (1 MeV equivalent n)/cm². The post-radiation tests performed on DB6-0 demonstrated the radiation tolerance of the KU FPGAs up to the exposed fluence. In the case of the ProASIC, some after-effects could be seen where the firmware was not working as expected and re-configuration was not possible. However, the firmware functionality of all devices was recovered after annealing, with re-configuration capability recovered all of the devices. The same applies to the IS25LP128F FLASH memories, where after annealing some of the devices recovered the re-programming capabilities, with all the contents of the memory not being affected in any of the devices.

Four different variants of 10 Gbps multi-mode SFP+ devices were tested: AVAGO, CORETEK, FS-Europe and Direkttronik. The two AVAGO AFBR-709SMZ devices completely failed to pass the irradiation tests done up to 9×10^{12} (1 MeV equivalent n)/cm² fluence. One out of nine CORETEK CT-A000NPP-SB1L-D devices tested failed to function after irradiating to 9×10^{12} (1 MeV equivalent n)/cm² fluence. In the case of the Direkttronik 33-4248 and the FS-Europe